

Polynomials 2 – Multiple Zeroes

1. The polynomial $x^3 - x^2 - 5x - 3$ has a double root at $x = \alpha$. What is the value of α ?
2. If $P(x) = x^4 - 4x^3 + 6x^2 - 4x + 1$, then:
 - a. Show that $x = 1$ is a zero of multiplicity 4
 - b. Fully factorise $P(x)$
3. If $P(x) = x^4 + x^3 - 3x^2 - 5x - 2$, then:
 - a. Show that $P(x)$ has a zero of multiplicity 3
 - b. Fully factorise $P(x)$
 - c. Solve the equation $P(x) = 0$
4. If $P(x) = x^3 - x^2 - 8x + 12$, then:
 - a. Show that $P(x)$ has a zero of multiplicity 2
 - b. Fully factorise $P(x)$
 - c. Solve the equation $P(x) = 0$
5. If $P(x) = 3x^4 - 8x^3 + 6x^2 - 24x + 40$, then:
 - a. Show that $P(x)$ has a zero of multiplicity 2
 - b. Fully factorise $P(x)$
 - c. Solve the equation $P(x) = 0$
6. The polynomial $P(x) = ax^4 + bx^3 + cx^2 + e$ has remainder -4 when divided by $(x + 1)$. The polynomial has a double root at $x = 1$.
 - a. Show that $2a + c = -3$
 - b. Hence, or otherwise, find the slope of the tangent to the graph $y = P(x)$ where $x = -1$
7. Let $P(x) = x^3 + ax^2 + bx - 4$, where a and b are real numbers. If $(x - 2)^2$ is a factor of $P(x)$, find the values of a and b .
8. The polynomial $P(s) = ax^3 + bx + c$ has a multiple zero at -1 and has remainder 8 when divided by $(x - 1)$. If a, b and c are real, find their values.
9. Solve each equation using the properties of polynomials.
 - a. $4x^3 - 8x^2 + 5x - 1 = 0$, given that it has a root of multiplicity 2.
 - b. $x^4 + 4x^3 - 16x - 16 = 0$, given that it has a root of multiplicity 3.
10. Solve each equation using the properties of polynomials.
 - a. $9x^3 + 12x^2 - 11x + 2 = 0$, given that it has a root of multiplicity 2.
 - b. $2x^4 + 5x^3 + 3x^2 - x - 1 = 0$, given that it has a root of multiplicity 3.
11. Consider the polynomial $P(x) = x^3 - 4x^2 - 3x + 18$.
 - a. i. Show that $P(3)$ and $P'(3)$ are both zero
ii. What can be deduced from the results in part i
 - b. Use part a. and the sum of zeroes to find all the zeroes of $P(x)$.
 - c. Hence factor $P(x)$
12. Consider the polynomial $P(x) = x^4 + 8x^3 + 18x^2 + 16x + 5$.
 - a. i. Show that $P(-1), P'(-1)$ and $P''(-1)$ are all zero.
ii. What can be deduced from the results in part i?
 - b. Use part a. and the product of zeroes to find all the zeroes of $P(x)$
 - c. Hence factor $P(x)$.
13. The polynomial $P(x) = x^3 - 27x + 54$ has a double zero.
 - a. Find the zeroes of $P'(x)$.
 - b. Determine which of the zeroes of $P'(x)$ is the double zero of $P(x)$.
 - c. Find the remaining simple zero of $P(x)$
14. The polynomial $P(x) = x^4 + 5x^3 - 75x^2 - 625x - 1250$ has a triple zero.
 - a. Find the zeros of $P''(x)$.
 - b. Determine which of the zeroes of $P''(x)$ is the triple zero of $P(x)$.
 - c. Find the remaining simple zero of $P(x)$.

15. The polynomial $P(x) = 2x^3 + 5x^2 - 4x - 12$ has a double zero.
 - a. Find the double zero
 - b. Find the remaining simple zero, and hence factor $P(x)$.
16. The polynomial $P(x) = 8x^4 - 28x^3 + 30x^2 - 13x + 2$ has a triple zero.
 - a. Find the triple zero
 - b. Find the remaining simple zero, and hence factor $P(x)$.
17. Consider the polynomial equation $x^4 - 10x^3 + 34x^2 - 42x + 9 = 0$.
 - a. Show that $x = 3$ is a double root of the equation.
 - b. Hence solve the equation.
18. The polynomial $P(x) = x^3 - 3x^2 - 9x + k$ has a double zero.
 - a. Find the two possible values of k .
 - b. For each of the possible values of k , factor $P(x)$.
19. The coefficients of the polynomial $P(x) = ax^3 + bx + c$ are real and $P(x)$ has a multiple zero at $x = 1$. When $P(x)$ is divided by $x + 1$ the remainder is 4. Find the values of a, b and c .
20. The polynomial $P(x) = x^4 + 7x^3 + 9x^2 - 27x + c$ has a triple zero.
 - a. Determine the value of the triple zero.
 - b. Hence find the value of c .
 - c. Factor $P(x)$.
21. a. Find the values of b and c if $x = 1$ is a double root of the equation $x^4 + bx^3 + cx^2 - 5x + 1 = 0$.
 b. Find the other roots of the equation.
22. It is known that $(x - 1)^2$ is a factor of the polynomial $P(x) = ax^{n+1} + bx^n + 1$. Show that $a = n$ and $b = -(1 + n)$.
23. The equation $Ax^3 + Bx^2 + D = 0$, where A, B and D are non-zero, has a double root.
 - a. Show that the double root is $-\frac{2B}{3A}$.
 - b. Hence show that $27A^2D + 4B^3 = 0$.
24. Prove that $P(x) = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots + \frac{x^n}{n!}$, where $n \geq 2$, has no multiple zeroes.
25. Let $P(x) = x^4 + mx^2 + n = 0$, where m and n are non-zero.
 - a. Prove that the equation $P(x) = 0$ cannot have a root of multiplicity greater than 2.
 - b. Let $x = a$ be a double root of $P(x) = 0$.
 - i. Prove that $x = -a$ is also a double root.
 - ii. What range of values can m take?
 - iii. Prove that $n = \frac{1}{4}m^2$, and write down the roots of the equation in terms of m .
26. The polynomial $P(x) = x^3 + 3px^2 + 3qx + r$ has a double zero.
 - a. Prove that the double zero is $a = \frac{pq-r}{2(q-p^2)}$.
 - b. Hence, or otherwise, prove that $4(p^2 - q)(q^2 - pr) = (pq - r)^2$.

Answer

1. $\alpha = -1$
2. b. $P(x) = (x - 1)^4$
3. b. $P(x) = (x + 1)^3(x - 2)$; c. $x = -1, 2$
4. b. $P(x) = (x - 2)^2(x + 3)$; c. $x = 2, -3$
5. b. $P(x) = (x - 2)^2(3x^2 + 4x + 10)$; c. $x = 2$
6. b. 12
7. $a = -5, b = 8$
8. $a = -2, b = 6, c = 4$
9. a. $x = \frac{1}{2}, \frac{1}{2}, -1$; b. $x = -2, -2, -2, 2$
10. a. $x = \frac{1}{3}, \frac{1}{3}, -2$; b. $x = -1, -1, -1, \frac{1}{2}$
11. b. 3, 3, -2; c. $P(x) = (x - 3)^2(x + 2)$
12. a. ii. -1 is at least a triple zero of $P(x)$;
 b. -1, -1, -1, 5; c. $P(x) = (x + 1)^3(x + 5)$
13. a. -3 and 3; b. 3; c. -6
14. a. $\frac{5}{2}$ and -5; b. -5; c. 10
15. a. -2; b. $\frac{3}{2}, P(x) = (x + 2)^2(2x - 3)$
16. a. $\frac{1}{2}$; b. 2, $P(x) = (2x - 1)^3(x - 2)$
17. b. $x = 3, 2 + \sqrt{3}$ or $2 - \sqrt{3}$
18. a. $k = 27$ or -5 ;
 b. When $k = 27, P(x) = (x - 3)^2(x + 3)$ and when $k = -5, P(x) = (x + 1)^2(x - 5)$.
19. $a = 1, b = -3, c = 2$
20. a. -3; b. $c = -54$; c. $P(x) = (x + 3)^3(x - 2)$
21. a. $b = -5$ and $c = 8$; b. $x = \frac{1}{2}(3 - \sqrt{5})$ or $\frac{1}{2}(3 + \sqrt{5})$
25. b. ii. $m < 0$, iii. $x = -\sqrt{-\frac{m}{2}}$ or $\sqrt{-\frac{m}{2}}$